

Emotional Persona: Real-Person Emotion Transfer for 3D Talking Avatars

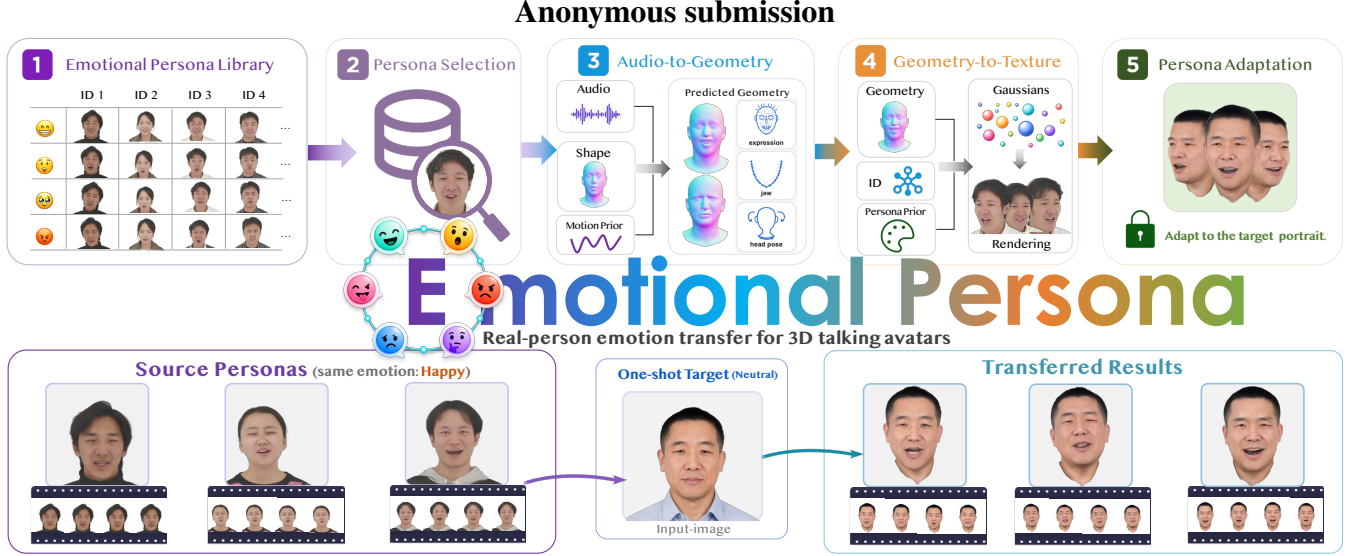


Figure 1: **Overview of Emotional Persona.** Given an emotional persona from the identity–emotion library, our framework transfers its person-specific emotional dynamics to an unseen one-shot avatar.

Abstract

Natural interaction with affective robots and embodied agents requires digital facial behavior that is both emotionally plausible and person-specific. Existing talking-head methods typically represent emotion using categorical labels or generic expression patterns, making it difficult to reproduce how a particular person expresses emotion. This challenge is especially pronounced for high-fidelity 3D talking avatars, where multi-view emotional data are scarce and person-specific dynamics must be transferred to one-shot target portraits. We introduce emotional personas, reusable priors derived from real multi-view performances that encode person-specific affective dynamics, and construct an Emotional Persona Library organized by identity and emotion. Building on this library, we present Emotional Persona, a unified 3D Gaussian avatar framework combining a persona-conditioned Audio-to-Geometry model, a dual-branch Geometry-to-Texture model, and one-shot Persona Adaptation. The framework animates an unseen one-shot portrait using the expression, jaw, and head-pose dynamics of a selected persona. We evaluate person-specific emotion transfer through reference-based perceptual and geometric measures and a blinded pairwise user study. Experiments demonstrate that our method achieves leading visual fidelity and motion accuracy while more faithfully preserving person-specific emotional dynamics than existing methods.

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Introduction

Recent portrait-animation and video-generation methods can synthesize visually compelling talking faces from a single image and lightweight control signals (Guo et al. 2024; Drobyshev, Bigata Casademunt, and Vougioukas 2024; Xu et al. 2025). Emotional behavior is commonly specified using categorical labels, learned emotion codes, or short reference motions (Peng et al. 2023; Daněček et al. 2023; Gan et al. 2023). While effective at producing recognizable emotions or speaking styles, most existing approaches do not explicitly aim to preserve the characteristic way in which a particular real person expresses emotion, including characteristic mouth asymmetry, brow dynamics, head-pose rhythm, and expression intensity. Capturing such person-specific behavior is important for affective agents, digital humans, film production, and social telepresence.

Existing evaluation protocols likewise focus primarily on visual fidelity, lip synchronization, and emotion classification (Ji et al. 2022; He et al. 2024; Tan et al. 2025). These criteria may reward a plausible or exaggerated generic expression even when it departs from the reference performer’s characteristic dynamics. Evaluating real-person emotion transfer instead requires reference-conditioned comparisons between the generated result and an audio-aligned performance at perceptual, geometric, and human-judgment levels. Fig. 1 illustrates this persona-transfer setting.

We therefore formulate real-person emotion transfer as follows: given an emotional persona and an unseen target portrait, animate the target with the persona’s characteris-

tic expression, jaw, and head-pose dynamics. This formulation shifts emotional control from category-level labels to reusable person-level references.

To realize this formulation, we construct an Emotional Persona Library from an expanded version of the EmoTalk3D dataset (He et al. 2024) and develop a three-stage avatar pipeline based on 3D Gaussian Splatting (Kerbl et al. 2023). The Audio-to-Geometry model predicts expression, jaw, and head-pose trajectories from speech; the dual-branch Geometry-to-Texture model maps predicted geometry and persona conditioning to Gaussian attributes; and Persona Adaptation specializes the library-pretrained avatar prior to an unseen target from a single portrait.

We evaluate person-specific emotion transfer using reference-conditioned perceptual and geometric measures, complemented by a blinded pairwise user study. Results demonstrate improved visual fidelity, motion accuracy, and preservation of person-specific emotional dynamics over existing methods. Our contributions are threefold:

- We introduce *emotional personas*, reusable person-level affective priors derived from captured multi-view performances, replacing categorical emotion labels with references that encode how a particular person expresses emotion.
- We present the first unified 3D Gaussian framework for transferring person-specific emotional dynamics from real performers to unseen one-shot avatars, combining Audio-to-Geometry, Geometry-to-Texture, and Persona Adaptation.
- We expand EmoTalk3D from 30 to 98 subjects with 68 synchronized multi-view captures and evaluate real-person emotion transfer using complementary reference-conditioned perceptual, geometric, and human assessments.

Related Work

Audio-Driven 3D Facial Animation

Audio-driven 3D facial animation maps speech signals to dynamic facial geometry. Early studies employed convolutional networks (Karras et al. 2017), recurrent architectures (Pham, Cheung, and Pavlovic 2017; Zhou et al. 2018), and subject-specific representations (Cudeiro et al. 2019). Subsequent methods improved temporal modeling and motion realism using transformers and learned motion priors. FaceFormer (Fan et al. 2022) formulated facial animation as autoregressive sequence generation, while CodeTalker (Xing et al. 2023) introduced discrete motion priors to alleviate over-smoothing. MeshTalk (Richard et al. 2021) and FaceXHuBERT (Haque and Yumak 2023) further improved audio-motion representations. Diffusion-based methods, including FaceDiffuser (Stan, Haque, and Yumak 2023), FaceTalk (Aneja et al. 2024), and DiffPoseTalk (Sun et al. 2024), generate diverse facial motions, whereas Imitator (Thambiraja et al. 2023) adapts motion to individual speaking styles. This line of work primarily focuses on speech articulation and geometric motion, rather than transferring person-specific emotional dynamics and their associated appearance variations.

Emotion-Aware Talking-Head Generation

Emotion-aware methods extend speech-driven animation with explicit affective control. In 3D, EmoTalk (Peng et al. 2023) and EMOTE (Daněček et al. 2023) disentangle emotion from speech-related motion, while EMOCA (Daněček, Black, and Bolkart 2022) introduces emotion-consistency objectives for expressive face reconstruction. ECAvatar (Yu et al. 2024b) improves expressive avatar rendering through detailed facial and oral modeling. StyleTalk (Ma et al. 2023) and StyleTalk++ (Wang et al. 2024) extract speaking and expression styles from reference videos.

In 2D portrait animation, EVP (Ji et al. 2021), EAT (Gan et al. 2023), and DICE-Talk (Tan et al. 2025) model emotional motion through cross-modal disentanglement or diffusion. Recent portrait video models, including EchoMimic (Meng et al. 2025b,a) and Hallo (Cui et al. 2025b,a), further improve visual fidelity and motion diversity. However, existing methods commonly represent emotion using categorical labels, manually specified intensities, or unconstrained latent style codes. Consequently, reproducing the characteristic temporal dynamics of a particular real performer is generally not an explicit objective and is rarely evaluated.

Gaussian Head Avatars and Real-Person Emotion Transfer

3D Gaussian Splatting (Kerbl et al. 2023) has enabled efficient and photorealistic head-avatar reconstruction. GaussianAvatars (Qian et al. 2024) and HeadGAP (Zheng, Wen, and Li 2024) develop animatable Gaussian representations, while recent systems such as Avat3r and Fast-GHA investigate data-efficient or generalizable reconstruction (Kirschstein, Romero, and Sevastopolsky 2025; Ji, Weiss, and Kansy 2026; Peng, Su, and Wang 2025; Wu, Zhou, and Hu 2026). Audio-driven Gaussian methods, including GaussianTalker, TalkingGaussian, EmoTalk3D, InsTaG, and SoulX-FlashHead (Yu et al. 2024a; Li et al. 2024; He et al. 2024; Li et al. 2025a; Yu et al. 2026), further improve rendering quality, synchronization, and animation control.

Reference-based portrait methods, such as EDTalk, EMO-Portraits, LivePortrait, and PersonaLive (Tan, Ji, and Bi 2024; Tan and Ji 2025; Drobyshev, Bigata Casademunt, and Vougioukas 2024; Guo et al. 2024; Xie, Xu, and Song 2024; Li et al. 2025b), provide strong motion and appearance priors. Single-image 3D methods such as VASA-3D and MimicTalk (Xu et al. 2025; Ye et al. 2024b,a) are closely related to our one-shot Persona Adaptation setting, while SadTalker and GeneFace++ (Zhang et al. 2023; Ye et al. 2023) provide representative audio-to-3DMM formulations for A2G. Across these lines of work, existing methods primarily target subject reconstruction, generic motion transfer, or emotion-label control. Complementing them, we model emotion through reusable persona priors captured from real performances rather than discrete conditions or unconstrained styles. We transfer the selected persona’s characteristic dynamics to a one-shot target and assess preservation of the reference performer’s emotional behavior using perceptual, geometric, and human-preference measures.

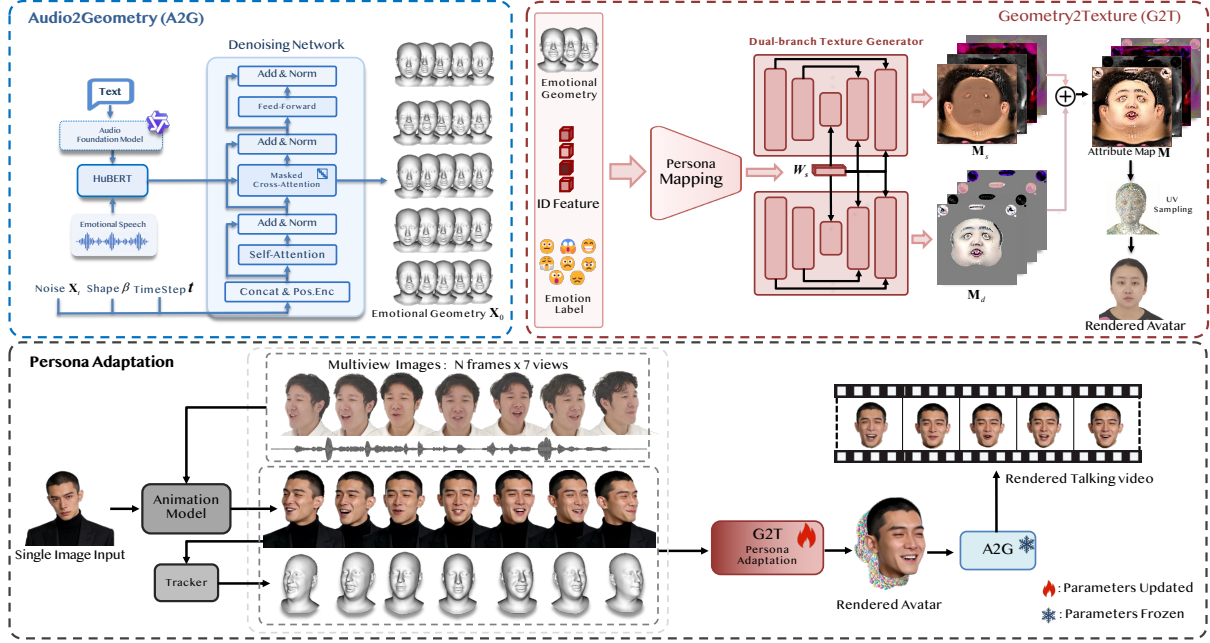


Figure 2: Overview of our pipeline. **Top left:** A2G predicts 3D emotional facial geometry from speech. **Top right:** A dual-branch G2T maps the predicted geometry to dynamic 3D Gaussian attributes. **Bottom:** Persona Adaptation specializes the library-pretrained G2T prior to a one-shot target, which is subsequently animated using the A2G predictions.

Method

Emotional Persona and Transfer Formulation

For identity i and emotion e , we define an emotional persona as

$$\mathcal{E}_{i,e} = \{\mathbf{A}_{i,e}, \Psi_{i,e}, \Theta_{i,e}, \mathbf{H}_{i,e}, \mathbf{V}_{i,e}, \beta_i\}, \quad (1)$$

where $\mathbf{A}_{i,e}$ is emotional speech, $\Psi_{i,e}$ and $\Theta_{i,e}$ are tracked FLAME (Li et al. 2017) expression and jaw-pose trajectories, $\mathbf{H}_{i,e}$ contains head rotation and translation, $\mathbf{V}_{i,e}$ contains the synchronized multi-view observations, and β_i is the identity shape.

An emotional persona is a structured real-person reference rather than a separately encoded latent code. Its speech and motion provide training data for A2G, while its tracked geometry and multi-view observations supervise G2T. Multiple identities are retained for each emotion, allowing the library to model person-specific expression, jaw, head-motion, and dynamic-appearance patterns beyond a shared category-level template.

Given a selected persona $\mathcal{E}_{s,e}$ and a one-shot target portrait $\mathbf{I}_w$, our goal is to animate the target using speech associated with the selected persona. The speech may be taken directly from the reference performance or synthesized in the same vocal and emotional style. The selected emotion category provides auxiliary appearance conditioning, while the target shape and appearance are estimated from $\mathbf{I}_w$ and remain fixed during animation.

Audio-to-Geometry

Given a driving audio sequence $\mathbf{A}$ of length T and target shape β_w , A2G predicts expression, jaw, and head-pose tra-

jectories:

$$[\hat{\Psi}, \hat{\Theta}, \hat{\mathbf{H}}] = \mathcal{M}_{\text{A2G}}(\mathbf{A}, \beta_w). \quad (2)$$

During persona transfer, $\mathbf{A}$ is the selected persona’s speech, whose vocal and emotional characteristics drive motion, while β_w constrains it to the target facial shape.

We employ a transformer-based diffusion model to generate the clean motion sequence $\mathbf{X}_0 = [\Psi, \Theta, \mathbf{H}]$. At diffusion step τ , the noisy motion $\mathbf{X}_\tau$ is combined with a condition token encoding τ and β_w , historical motion context, and positional encoding. Stacked transformer decoder blocks model temporal motion dependencies through self-attention and attend to temporally aligned HuBERT audio features through masked cross-attention.

During inference, classifier-free guidance combines conditional and unconditional noise predictions:

$$\tilde{\epsilon} = \epsilon_\theta(\mathbf{X}_\tau, \emptyset) + w [\epsilon_\theta(\mathbf{X}_\tau, \mathbf{A}, \beta_w) - \epsilon_\theta(\mathbf{X}_\tau, \emptyset)]. \quad (3)$$

The clean expression, jaw, and head-pose trajectories are recovered according to the diffusion schedule.

A2G is optimized with vertex reconstruction, first-order velocity, and second-order smoothness objectives:

$$\mathcal{L}_{\text{A2G}} = \lambda_{\text{vert}} \mathcal{L}_{\text{vert}} + \lambda_{\text{vel}} \mathcal{L}_{\text{vel}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}}. \quad (4)$$

The three terms respectively constrain geometric accuracy, motion velocity, and temporal stability.

Geometry-to-Texture

G2T converts the motion predicted by A2G into dynamic 3D Gaussian attributes. We extract the target identity feature $\mathbf{F}_w$ from the input portrait using DINOv2 (Oquab et al. 2023).

At frame t , a persona mapping network combines $\mathbf{F}_w$, the predicted expression and jaw pose, and the selected persona’s one-hot emotion label $\mathbf{L}_e$:

$$\mathbf{w}_t = \mathcal{F}_{\text{map}}(\mathbf{F}_w, \hat{\psi}_t, \hat{\theta}_t, \mathbf{L}_e). \quad (5)$$

The emotion label is used only as auxiliary appearance conditioning; the person-specific temporal behavior is carried by the speech-driven geometry predicted by A2G.

The mapping network fuses identity, motion, and emotion features into the style representation $\mathbf{w}_t$, which modulates a dual-branch texture generator. The two branches predict a static Gaussian attribute map $\mathbf{M}^s$ and a motion-dependent residual map $\mathbf{M}_t^d$:

$$\mathbf{M}_t = \mathbf{M}^s + \mathbf{M}_t^d. \quad (6)$$

The static branch models the target’s base appearance, while the dynamic branch captures expression-dependent changes. Both branches are implemented with StyleUNets whose synthesis blocks are modulated by $\mathbf{w}_t$. Lateral encoder features preserve the FLAME UV structure throughout the synthesis process.

Each pixel in $\mathbf{M}_t$ stores the position, rotation, scale, opacity, and spherical-harmonic coefficients of a 3D Gaussian primitive. Fixed barycentric coordinates sample these attributes on the deformed FLAME surface, producing the dynamic Gaussian set

$$\mathcal{G}_t = \mathcal{M}_{\text{G2T}}(\hat{\psi}_t, \hat{\theta}_t, \mathbf{F}_w, \mathbf{L}_e). \quad (7)$$

The predicted head pose $\hat{\mathbf{H}}_t$ is applied as a global rigid transformation during rendering.

G2T is trained with reconstruction and attribute-regularization objectives:

$$\mathcal{L}_{\text{G2T}} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{static}} + \lambda_{\text{dyn}} \mathcal{L}_{\text{dynamic}}. \quad (8)$$

Here, $\mathcal{L}_{\text{rec}}$ combines photometric and perceptual losses, while the remaining terms regularize the static and dynamic Gaussian attribute maps.

Persona Adaptation

Persona Adaptation transfers the learned avatar prior to a one-shot target while preserving the selected persona’s emotional characteristics. Given a target portrait $\mathbf{I}_w$, we use a pretrained portrait animation model (X-NeMo (Zhao et al. 2025) by default) to synthesize sparse pseudo-views. Driven by speech from the selected persona $\mathcal{E}_{s,e}$, these observations exhibit corresponding expression and head-motion patterns. A tracker estimates their FLAME geometry, target shape β_w , camera parameters, and head poses.

Directly reconstructing an independent set of 3D Gaussians from these sparse observations is under-constrained. Instead, we adapt the G2T generator pretrained on the Emotional Persona Library, using its learned multi-view avatar prior to regularize the reconstruction.

Adaptation follows two-stage Pivotal Tuning Inversion (Roich et al. 2021). In the first stage, the G2T weights are frozen and a target pivot $\mathbf{w}_{\text{pivot}}$, initialized from the DINOv2 target feature, is optimized to reconstruct the pseudo-view

observations under their tracked geometry. In the second stage, the pivot is fixed and the G2T weights are tuned using all available pseudo-views. A locality term constrains the adapted generator to remain close to the pretrained prior:

$$\mathcal{L}_{\text{PTI}} = \mathcal{L}_{\text{rec}} + \lambda_{\text{reg}} \mathcal{L}_{\text{loc}}. \quad (9)$$

After adaptation, the target pivot, shape, and base appearance remain fixed. A2G predicts expression, jaw, and head-pose trajectories from the selected persona’s speech, while the adapted G2T maps the predicted geometry to dynamic Gaussian attributes. Persona speech and induced pseudo-view motion jointly condition adaptation. Network configurations, losses, diffusion settings, and PTI details are provided in the supplement.

Experiments

Experimental Setup

Dataset. We construct an upgraded version of EmoTalk3D (He et al. 2024), expanding it from 30 to 98 subjects and improving the acquisition pipeline through hardware-synchronized high-resolution multi-view capture, calibrated color reproduction, stable illumination, and synchronized audio. The resulting dataset covers eight emotions with phonetically balanced speech, and each sequence is tracked with VHAP (Qian 2024) to recover FLAME motion, identity shape, and camera parameters. All identities are used to train the Emotional Persona Library. Further details of the capture system, calibration, and data collection are provided in the supplementary material.

Implementation Details. For all main comparisons, our method uses X-NeMo (Zhao et al. 2025) to generate the pseudo-view observations; we vary this prior only in the animation-prior ablation. A2G and G2T are optimized with Adam/AdamW. PTI takes less than 10 minutes per target on one RTX A6000, after which rendering exceeds 200 FPS. Additional training details are provided in the supplementary material.

Evaluation Metrics. All quantitative comparisons use the frontal view at 512×512 and 25 FPS. Frame-based metrics use all valid frames, FVD uses non-overlapping 16-frame clips, and side views are evaluated qualitatively. We report PSNR, SSIM, LPIPS, FID, FVD, LMD, AUE, SyncNet, and Emo-Score (Ryumina, Dresvyanskiy, and Karpov 2022) following their standard definitions. To measure preservation of a selected real-person reference, we define VLM Reference Similarity (VRS) and Expression Trajectory Distance (ETD). For $K = 8$ audio-aligned frame pairs rated by a fixed VLM on a five-point scale,

$$\text{VRS} = \frac{1}{K} \sum_{k=1}^K \frac{s_k - 1}{4}, \quad s_k \in \{1, \dots, 5\}. \quad (10)$$

For aligned FLAME expression trajectories of length T ,

$$\text{ETD} = \frac{1}{T} \sum_{t=1}^T \left\| \hat{\psi}_t - \psi_t^{\text{ref}} \right\|_2. \quad (11)$$

Higher VRS and lower ETD indicate closer agreement with the reference persona. We further conduct a blinded pairwise user study to directly evaluate whether the generated

Method	Visual Fidelity			Motion Quality					Emotion Recognition			Persona Transfer	
	PSNR $\uparrow$	LPIPS $\downarrow$	SSIM $\uparrow$	FVD $\downarrow$	FID $\downarrow$	LMD $\downarrow$	AUE $\downarrow$	Sync $\uparrow$	Emo. $\uparrow$	Neu. $\uparrow$	Non-neu. $\uparrow$	VRS $\uparrow$	ETD $\downarrow$
EAMM (Ji et al. 2022)	9.822	0.405	0.575	941.371	123.278	20.253	4.833	2.146	0.261	0.339	0.217	0.421	0.236
EAT (Gan et al. 2023)	14.713	0.321	0.631	605.731	110.323	14.825	4.252	3.121	0.432	0.413	0.464	0.552	0.188
GaussianTalker (Yu et al. 2024a)	26.931	0.154	0.776	512.829	73.710	3.616	3.631	5.762	0.313	0.482	0.251	0.574	0.174
TalkingGaussian (Li et al. 2024)	27.515	0.175	0.793	508.387	78.192	3.592	3.715	6.012	0.316	0.536	0.277	0.583	0.171
EmoTalk3D (He et al. 2024)	22.372	0.152	0.837	467.774	76.320	3.563	3.576	4.173	0.371	0.485	0.353	0.622	0.149
InsTaG (Li et al. 2025a)	28.114	0.098	0.838	498.332	71.897	3.312	3.591	6.325	0.334	0.523	0.314	0.631	0.143
DICE-Talk (Tan et al. 2025)	26.831	0.214	0.779	376.024	54.441	3.411	3.435	6.267	0.481	0.514	0.530	0.596	0.168
SoulX-FlashHead (Yu et al. 2026)	27.317	0.172	0.811	391.119	56.182	3.392	3.475	5.901	0.338	0.541	0.511	0.654	0.132
Ours	30.037	0.081	0.892	367.882	53.199	3.236	3.212	5.571	0.452	0.551	0.427	0.781	0.097

Table 1: Quantitative comparison with existing methods.



Figure 3: Frontal-view qualitative comparison. Results for two subjects with “happy” and “sad” emotions.



Figure 4: **Side-view qualitative comparison.** Results for two subjects with “sad” and “happy” emotions.

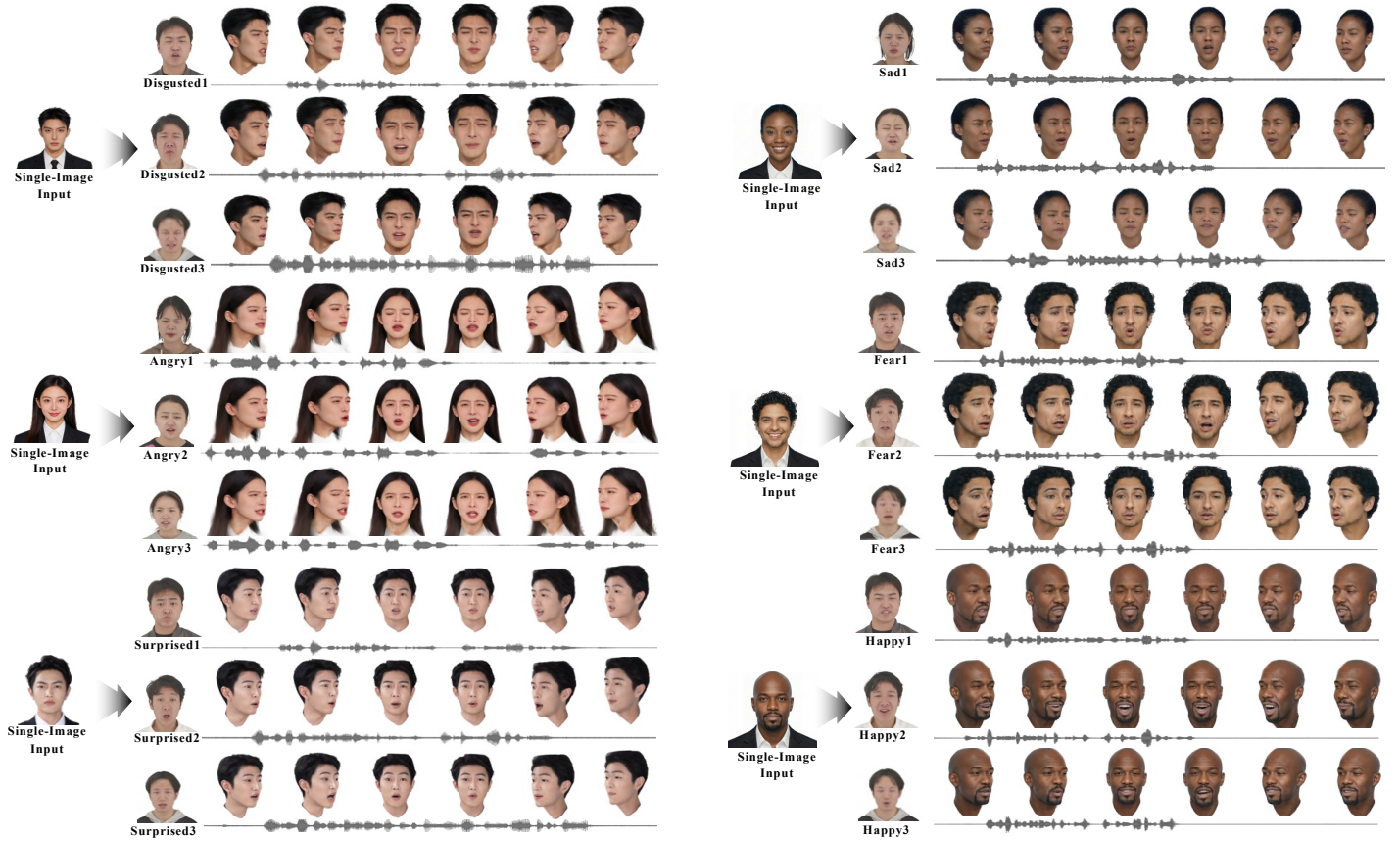


Figure 5: **Persona-conditioned emotion transfer.** For each emotion, three source personas animate a shared one-shot target (left), producing distinct person-specific expression and head-motion patterns beyond a category-level response.

(a) Audio-to-Geometry (A2G) Ablation						
Variant	FVD↓	FID↓	LMD↓	AUE↓	Sync↑	Emo↑
Unseen emotion	603.733	72.278	4.012	4.293	<u>5.517</u>	0.301
w/o $\mathcal{L}_{smooth}$	402.910	63.981	3.790	3.309	2.733	0.397
w/o $\mathcal{L}_{vel}$	418.113	67.856	3.872	3.688	3.762	<u>0.446</u>
Full model	367.882	53.199	3.236	3.212	5.571	0.452

(b) Geometry-to-Texture (G2T) Ablation			
Variant	PSNR↑	LPIPS↓	SSIM↑
Static-only branch	28.371	0.108	0.762
w/o emotion fusion	28.638	0.096	0.793
w/o $\mathcal{L}_{static}$	29.073	0.093	0.855
w/o $\mathcal{L}_{dynamic}$	<u>29.382</u>	<u>0.090</u>	<u>0.873</u>
Full model	30.037	0.081	0.892

Table 2: Ablation Study on A2G and G2T Modules.

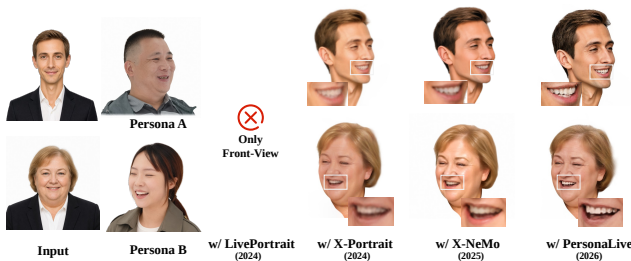


Figure 6: **Qualitative comparison of 2D animation priors.** LivePortrait produces only frontal animations; without side-view observations, 3D reconstruction fails.

results preserve the reference person’s emotional style. The VLM configuration, study protocol, and statistical analysis are provided in the supplementary material.

Ablation Study

Ablations of A2G and G2T. Table 2 presents ablations of the A2G objectives and G2T design. For A2G, *Unseen emotion* evaluates generalization to unseen emotional styles synthesized by Qwen3-TTS (Hu et al. 2026), whereas removing $\mathcal{L}_{vel}$ and $\mathcal{L}_{smooth}$ assesses the contributions of first-order motion consistency and second-order temporal stability, respectively. For G2T, *Static-only branch* replaces the dual-branch generator with a single static branch, *w/o emotion fusion* removes emotion conditioning, and the remaining variants remove the corresponding branch-specific objectives. The full model achieves the best overall performance. Bold and underlined values denote the best and second-best results, respectively.

Effect of the 2D Animation Prior. We vary only the pseudo-view animation prior. LivePortrait cannot synthesize the side views required for valid 3D reconstruction and is therefore excluded from the adapted quantitative comparison. PTI time and post-adaptation FPS are measured on one RTX A6000. Table 3 and Fig. 6 show that stronger multi-view priors improve view consistency, local detail, and adaptation ef-

Prior	PSNR↑	LPIPS↓	FID↓	Emo-Score↑	PTI (min)↓	FPS↑
LivePortrait	No view-consistent multi-view output (PTI N/A)					
X-Portrait	28.785	0.115	54.83	0.429	< 12	
X-NeMo	30.037	0.081	53.19	0.452	< 10	200+
PersonaLive	30.412	0.077	51.78	0.458	< 3	

Table 3: **Comparison of 2D animation priors.**

ficiency. X-NeMo is used by default, while the additional gains from PersonaLive indicate that advances in animation priors translate directly into improvements in our framework, providing a clear path for continued performance gains as stronger priors emerge.

Comparison with Existing Methods

Quantitative Comparison. All methods are evaluated on the frontal view under the common protocol described above. Dark and light purple in Table 1 indicate the best and second-best results. Single-image methods use the same target portrait and audio. EAMM additionally receives the persona clip; EAT receives the matched emotion and its required head-pose sequence; DICE-Talk receives the matched emotion label; and SoulX-FlashHead uses its official image-audio inputs. Following their official protocols, GaussianTalker and TalkingGaussian use frontal target video, EmoTalk3D synchronized multi-view data, and InsTaG a five-second target video. We therefore report a protocol-matched rather than equal-supervision comparison; reproduction details are provided in the supplementary material.

Table 1 shows that Emotional Persona achieves the best overall reconstruction and motion quality. Although DICE-Talk obtains a higher Emo-Score, our higher VRS and lower ETD indicate closer preservation of the selected persona’s emotional dynamics.

Qualitative Comparison. Figs. 3 and 4 present the frontal- and side-view comparisons, respectively. EAMM exhibits blurry textures; EAT and DICE-Talk produce exaggerated deformations that deviate from the subject’s natural style. The 3D-based methods improve rendering fidelity but can exhibit unnatural mouth shapes under large pose changes. Emotional Persona better balances visual fidelity and subject-specific emotional behavior.

Conclusion

We introduced Emotional Persona, a 3D Gaussian avatar framework that transfers person-specific emotional dynamics from real multi-view performances to one-shot target portraits. By combining an Emotional Persona Library with A2G, G2T, and one-shot Persona Adaptation, the framework moves emotional control beyond category labels toward reusable real-person references. Experiments demonstrate gains in visual fidelity, motion accuracy, and person-specific emotion transfer, offering a path toward more individualized and expressive digital avatars.

Limitations. FLAME may not fully capture extreme expressions or complex mouth geometry, leading to artifacts during persona adaptation, especially under large mouth openings.

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